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Appendix A — PT Interview Transcript

Interview Metadata

- Interviewer: Marwan Ashhuby
- Interviewee role: Qualified Personal Trainer (name withheld for anonymity: PT-1)
- Date of interview: 20/03/26
- Location: Audio call
- Format: Pre-written structured questions
- Duration (approximate in minutes): 28
- Recording method: Audio Record & transcribe (verbatim)

Background Questions

1. How long have you been working as a personal trainer?
   I have officially worked with clients since 2022, but I have been active at the gym since I was around 17 so 10 years now.

2. What types of clients do you typically train (beginners, intermediate, advanced)?
   Honestly, I get all types of people, I personally specialise in rehabilitation and physical therapy, so a lot of my clients will be older men and women anywhere from 30 to 80. Theres also a lot of athletes or amateur sports players who are perhaps injured and want help recovering so they can return to play. Apart from that, I get the regular clients that any other trainer might get [such as] beginners wanting to get into shape/ get healthier and so on.

3. Do you currently use any apps yourself for tracking client progress or your own training?
   I have used a few apps over the years; I'll tell you about the apps I have used as a trainer. In the past I have used Everfit, Trainerize and mypthub. They more of less do the same thing, but I have most recently switch from mypthub to Trainerize as [Trainerize] allows me to track client's nutrition as well as the obvious stuff like book sessions, manage clients, message them, automated check ins. These apps are made primarily for trainers, so I would not recommend it for clients especially as they can be quite costly. For myself, I used to use the typical gym apps everyone else uses, they all have their down sides but now I use Trainerize as I am already paying for it and comfortable with the layout, so might as well.

Problem Space Questions

4. In your experience, do your clients track their workouts? If so, how?

Yeah definitely, I'd say nearly all of them do. It's the most common question I get outside of 'how much protein should I be having?'. I normally say the same thing depending on what kind of exercises we are doing, you know? Strava for running, MyFitnessPal for diet, Shred is a good one for weightlifting if you don't mind paying it can be quite expensive. I think £15 a month, which you know is essentially the price of a gym membership. I think they offer a free plan, but it doesn't include the main features like progress tracking and exercises libraries. I think for beginners they get too hung up on which apps to use and which exercises to do, when really, they should just pick one and just start.

5. What are the most common problems you see clients having with tracking their training?

As mentioned, a lot of the best training apps out there are paid and when you are dealing with clients who [for them] training is not the most important thing ever in their life, it can be hard to justify another subscription especially with how things are right now. There are plenty of free alternatives and free plans and I think for most gym goers this will work well enough. They might have to download a couple free weightlifting apps like use Shred [free plan] for building a workout and Hevy [free plan] for tracking progress.

6. Have you ever recommended a specific app to a client? If so, which one?

Question skipped.
[Interviewer note: PT-1 addressed this question within their response to Q4 and Q5 above.]

7. Do you think anything is missing from the current fitness app landscape?

I think people kinda rush to the next big app a little too easy. They see a TikTok saying this app is great by their favourite influencer and instantly think they need to switch. I think it's a distraction especially with all the ai in these newer apps. I'm sure it can be really helpful for complete beginners who need extra guidance but to be I think apps need to be simple and easy to navigate and use and [that] they don't cause my phone to heat up after using it for 5 minutes. Theres so many apps made for the 'beginner' as they are the biggest market, I mean think about all the people who join around December and January, I'm sure these apps make a fortune around this time, but there seems to be such a skew towards apps for newer people with features intermediate or advanced lifters wont ever use and it's like really? is no one going to make a simple free app that the average gym goer will actually use and enjoy using.

Feature Validation Questions

8. How important do you think is the role of digital logging in a gym goers long-term progress compared to other methods (like paper or memory)?

I think its very important. When I first started out, I attempted to just remember every workout and weight, and it was a huge mess. Its really just not worth the risk, forgetting just one number can halt your progress especially if your memory is as bad as mine. Really workout logging is THE feature of weightlifting apps, if you don't have it then there is little point in pressing the download button. Machines don't forget.

9. How important is a rest timer that is integrated directly into the workout log, rather than a separate app?

I'd also say very important. Honestly, I think this is the number one feature 90% of gym apps missing especially the free to use ones. I'm not quite sure why it's such an oversight, maybe because they know you already have a timer on your phone, but then you have to go back and forth every single time you do a set. It's not a huge deal which I guess is why not more apps have added it, but yeah a very useful thing to have.

10. How useful would progress charts showing strength gains be for your clients? If those charts were available, what specific data would they want to see displayed?

Extremely useful, 99% of my clients get their motivation from seeing progress. If we get stuck on a weight for more than a month a lot of them, especially the newer gym goers, get disheartened and [it would be great if] they saw the progress and what they have already accomplished and see that graph go up and up, they get all confident and realise they are capable of doing so much more than they realise.

11. How important is it for your clients to be able to see their previous workouts if they log in from a different device?

I mean everyone loses their phone or upgrades every Christmas or whenever, so honestly, I would be really disappointed if I couldn't move my data onto the new phone. As I said seeing the progress chart and everything you accomplished is such a great feeling, I wouldn't want to lose all that data just because I'm switching phones.

12. How do you feel about turning elements of working out into a game (streaks, personal bests, badges) as motivational tools for gym-goers?

That's a really great idea, it's something I basically already do with clients. Because if you think about it the gym is basically a video game, a PR is just a high score. Earning achievements and badges or whatever would make you feel seen or like you are being appreciated for your accomplishments like 'this is your 50th day working out this year, good job', that's sure to make anybody feel great. People think my job is telling people what exercises to do and what machine to stand next to, but its really not. All these resources are online free to download, if that was the

case people would fire me after my first week with them after I teach them the basics. My job is encouragement, its keeping people motivated long enough where habit overtakes motivation, and they are now in the gym a few times a week because its Tuesday and Tuesday is push day. Streaks are obviously going to be really important for this too, I'm sure your aware consistency is the biggest hurdle for gym goers, even those who have been going for years. When people have a streak, they don't want to lose it. That might make people go even when they plan on skipping just to keep it alive.

Open/Close

13. Is there anything else you think a fitness tracking app absolutely must have to be genuinely useful?

As I said, workout logging is a must, unless it's a super specialised app you have to be able to log your workouts. I like your ideas about rest timers and being able to log on with a different device, I think these are a very underrated features that are excluded for more apps then you think. If you haven't already, presenting the workouts you logged as graphs is super helpful. Reading plain numbers is hard to visualise. I think those are the core features you really should have and omitting them would probably lead to a worse experience.

14. Are there any features in existing apps that you or your clients find difficult or frustrating to use?

Um, mostly it's the subscription models and free trials that annoy me. They [Clients] also ask me if its worth paying for these apps and I don't really know what to tell them half the time. Because if your new and this is your first day your not really going to use the features your paying for, but at the same time they make the free plans so barebones and borderline unusable, which makes sense because they want you to pay, but yeah if your going to make an app for the average weight lifter it really needs to be free with no ads if that's possible and your not looking to make money with your app.

Appendix B — Participant Information Sheet

Study Title: Evaluating the usability of a unified fitness tracking application

Invitation: You are being invited to take part in a short usability study as part of a final year university project. Please read this sheet before deciding whether to participate.

Purpose: This study aims to evaluate how easy and enjoyable a mobile fitness tracking application is to use. Your feedback will directly inform the final design evaluation.

What will happen: You will be asked to complete five short tasks on a mobile application. The session will take approximately 20–30 minutes. The researcher will observe and take notes. You may think aloud as you work — narrating what you are doing and why.

Voluntary participation: Your participation is entirely voluntary. You may stop at any time without giving a reason.

Confidentiality: No personally identifiable information will be recorded. You will be referred to only as P1, P2, etc. in the final report. No audio or video recording will be made without your explicit consent.

Data use: Anonymised responses will be used solely for the purposes of this academic project and will not be shared with third parties.

Researcher: Marwan Ashhuby, BSc Software Engineering, University of Hull.
Supervisor: John Whelan.

Questions: Contact the researcher before agreeing to participate if you have any concerns.

Appendix C — Consent Form Template

I confirm that I have read and understood the Participant Information Sheet.

I understand that my participation is voluntary and I may withdraw at any time without penalty.

I understand that my responses will be anonymised and used only for academic purposes.

I agree to take part in this study.

Name: _____

Signature: _____

Date: _____

Appendix D — Onboarding Google Form

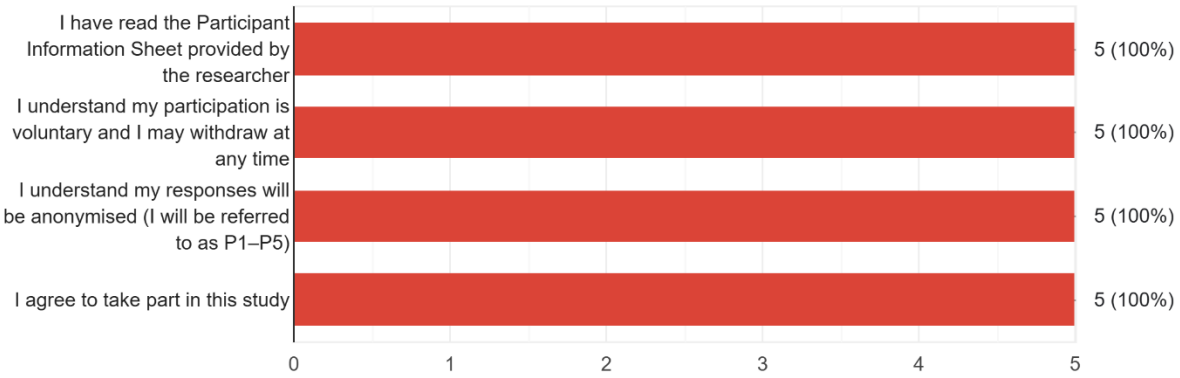
Section 1: Consent *

Before using the app, please confirm the following:

- [ ] I have read the Participant Information Sheet provided by the researcher
- [ ] I understand my participation is voluntary and I may withdraw at any time
- [ ] I understand my responses will be anonymised (I will be referred to as P1–P5)
- [ ] I agree to take part in this study

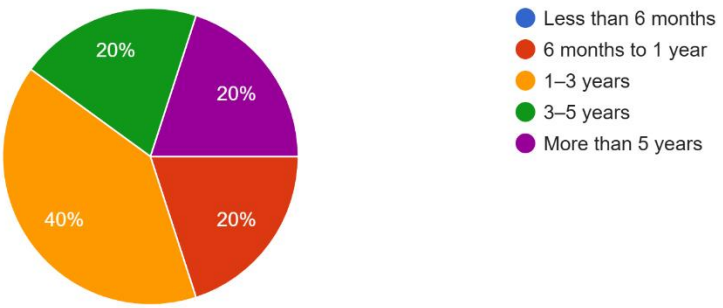
Section 1: Consent Before using the app, please confirm the following:

5 responses



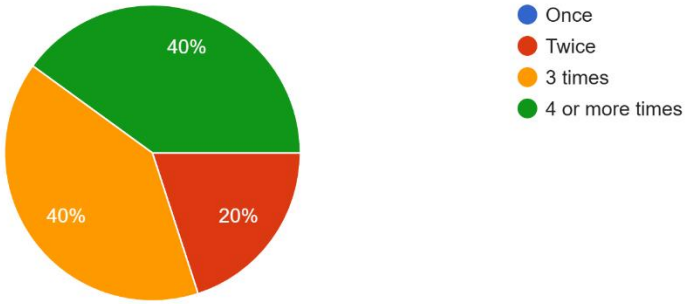
How long have you been going to the gym regularly?

5 responses



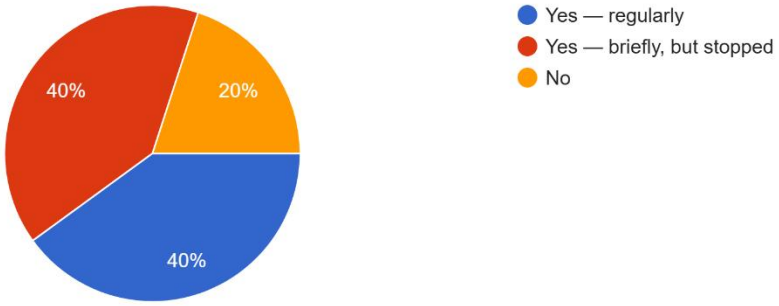
How often do you currently train per week?

5 responses



Have you used a fitness tracking app before?

5 responses



If yes, which app(s) have you used?
(leave blank if no)

4 responses

Shred

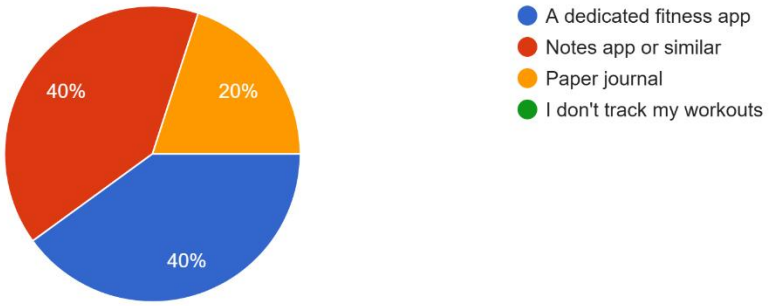
Hevy

Strava

Trainerize

What method do you currently use to track your workouts

5 responses



Section 4 of 4

Section 3: Instructions

Thank you. Here is what to do over the next 7 days:

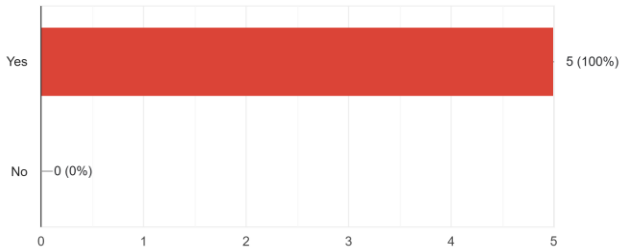
- Use OneRep during your actual gym sessions as you normally would. Log your exercises, sets, weights, and reps.
- Use the rest timer between sets.
- After each session, take 2 minutes to check your Progress screen.
- If anything confuses you, frustrates you, or impresses you — make a quick voice note or text note to yourself. You will be asked about these at the end.
- You will receive a short check-in form on Day 3. Please complete it — it takes under 3 minutes.
- At the end of Day 7, you will meet with the researcher in person for a 20–30 minute session.

Do you understand these instructions? *

- [ ] Yes
- [ ] No

Do you understand these instructions?

5 responses



Appendix E — Day 3 Check-In Form

OneRep — Day 3 Check-In

Don't worry this will only take two minutes to complete, Thank you :D

[redacted]

Not shared

* Indicates required question

Have you been able to use OneRep during your workouts so far? *

- ◯ Yes — for every session
- ◯ Yes — for some sessions
- ◯ No — I haven't had a gym session yet
- ◯ No — I had trouble with the app

On a scale of 1–5, how easy has the app been to use so far? *

	1	2	3	4	5	
Very difficult	◯	◯	◯	◯	◯	Very easy

Has anything confused or frustrated you so far?

Your answer

Has anything worked particularly well or surprised you positively?

Your answer

Is there anything you expected the app to do that it does not currently do?

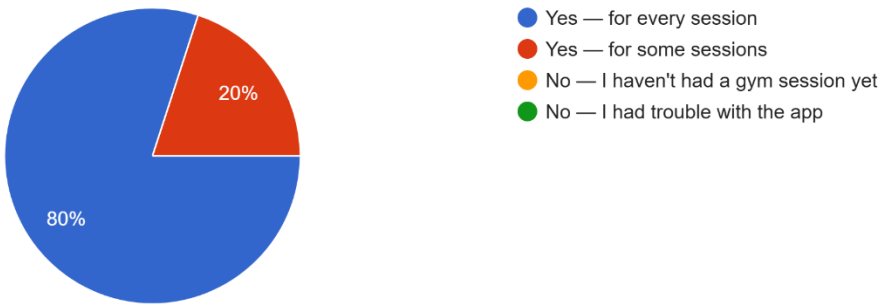
Your answer

Submit

Clear form

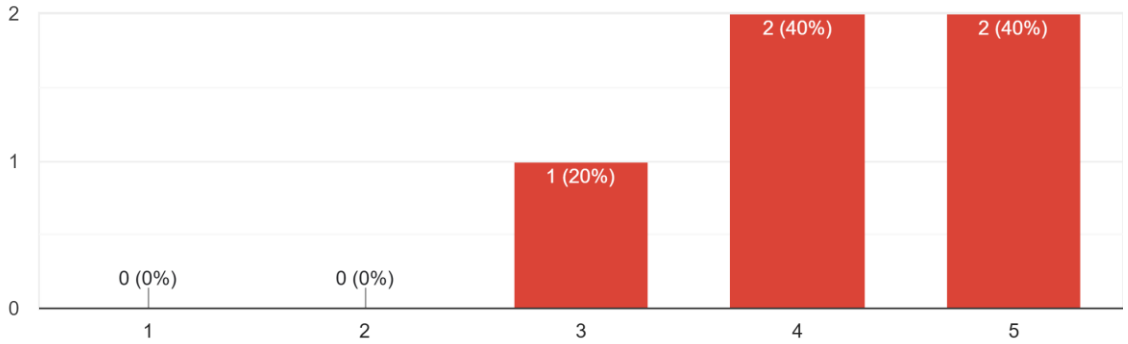
Have you been able to use OneRep during your workouts so far?

5 responses



On a scale of 1–5, how easy has the app been to use so far?

5 responses



Has anything confused or frustrated you so far?

5 responses

Would be really cool if it 'auto opened' onto your recent or most used split because having to pick the same one everyday felt unnecessary as most people only really use one 99% of the time.

I really wish it loaded your previous weight/reps or pr for any given exercise e.g. if I did 120kg on leg extensions last week that data should be filled in already.

I have to scroll and search abit when i add new exercises bc its a big list it rlly should be aphetically or sectioned by body part. Its not terrible because I only have to add exercises very very rarely but its still a little annoying.

No

its hard to navigate the exercise page

Has anything worked particularly well or surprised you positively?

5 responses

I got the 7 day badge at the end which was nice, I would love to shoot for 30 days if I get access to the app after it releases.

I love the freestyle session, its so so so helpful because I cant always commit to the split because im so so busy. So I have to do these mini sessions alot as i dont have the time to stay an hour/ hour and a half. Its so cool because normally if I do this it gets complicated and I use the notes to track which I cant do for freestyle sessions but with OneRep It carries over my progress and PRs and stuff.

I really like how it only takes two or three clicks from when I enter a gym to start a new session. Split to day to start. very simple

I really like how the time is automatic and buzzes your phone when its time to do another set it really stops me day dreaming for too long

I love the logging. I normally use a little notebook but honestly the ease of use with an app is unreal. The attendance tracking is also a huge plus.

Is there anything you expected the app to do that it does not currently do?

3 responses

Different colour palettes or modes, Let me customise it more. Widget dashboard where I can put all my favorite 'trophies' or PRs so i can see the number to beat when I open the app and other widgets like 'quick start' for common days like upper or lower so i can start with only one click.

Probably not what your looking for but being able to add friends and see pbs and ranking would be cool. I want to see top 5 bench press max among my friends because we are really competative about it.

no

Appendix F — Post-Session Questionnaire

OneRep — Post-Session Questionnaire

itsashhuby@gmail.com Switch account

Not shared

Section 1: System Usability Scale

Please rate each statement based on your overall experience with OneRep over the past week and today's session. Use a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I think that I would like to use this app frequently

1 2 3 4 5

I found the app unnecessarily complex

1 2 3 4 5

I thought the app was easy to use

1 2 3 4 5

I think that I would need the support of a technical person to use this app

1 2 3 4 5

I found the various functions in this app were well integrated

1 2 3 4 5

I thought there was too much inconsistency in this app

1	2	3	4	5
○	○	○	○	○

I would imagine that most people would learn to use this app very quickly

1	2	3	4	5
○	○	○	○	○

I found the app very cumbersome to us

1	2	3	4	5
○	○	○	○	○

I felt very confident using the app

1	2	3	4	5
○	○	○	○	○

I needed to learn a lot of things before I could get going with this app

1	2	3	4	5
○	○	○	○	○

Back Next Clear form

Section 2: Qualitative Feedback

What was the single most frustrating or confusing moment you experienced with the app, either today or during the week? If none, write "None".

Your answer

What was the single best or most useful feature of the app?

Your answer

Compared to any fitness apps you have used before, how does OneRep compare? *

- ◯ Much worse
- ◯ Slightly worse
- ◯ About the same
- ◯ Slightly better
- ◯ Much better
- ◯ I have not used other fitness apps

Would you continue using this app after this study? *

- ◯ Yes, definitely
- ◯ Probably yes
- ◯ Unsure
- ◯ Probably not
- ◯ No

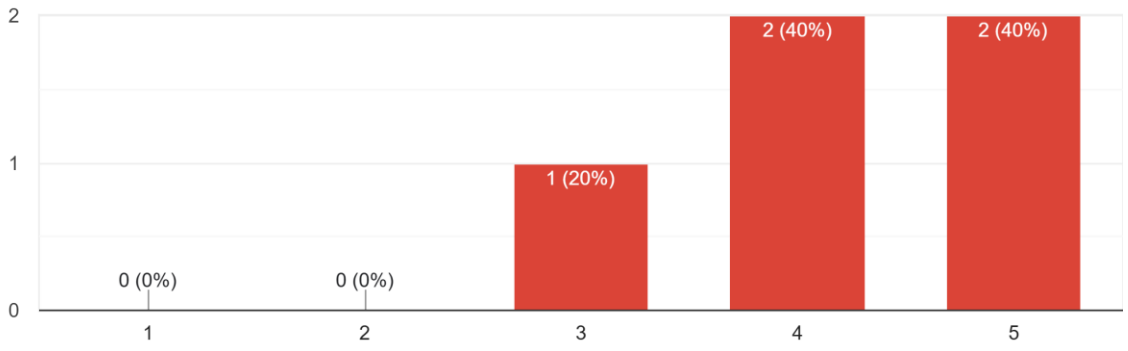
Any other comments or suggestions?

Your answer

Back | Submit | Clear form

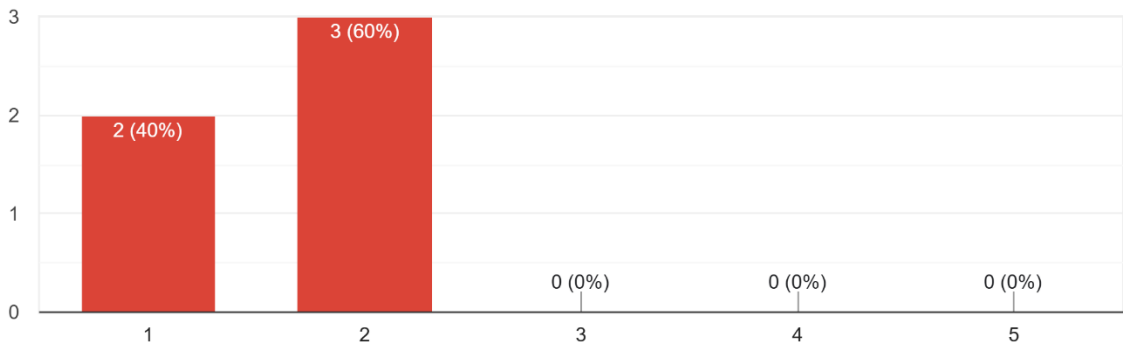
I think that I would like to use this app frequently

5 responses



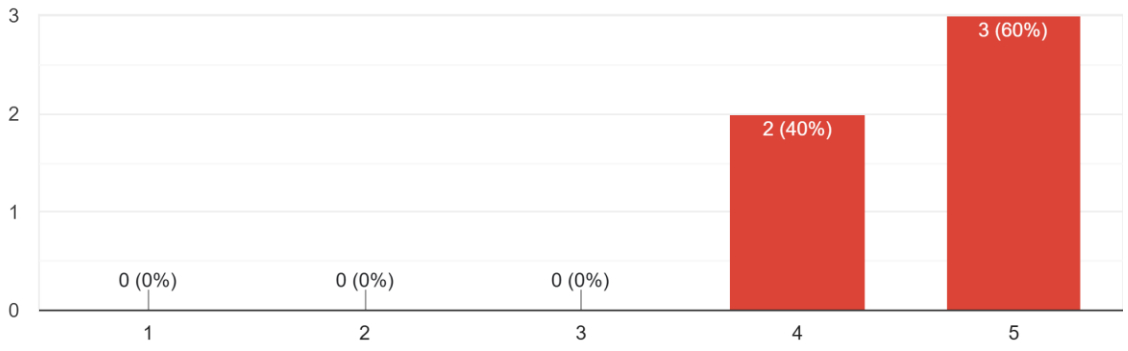
I found the app unnecessarily complex

5 responses



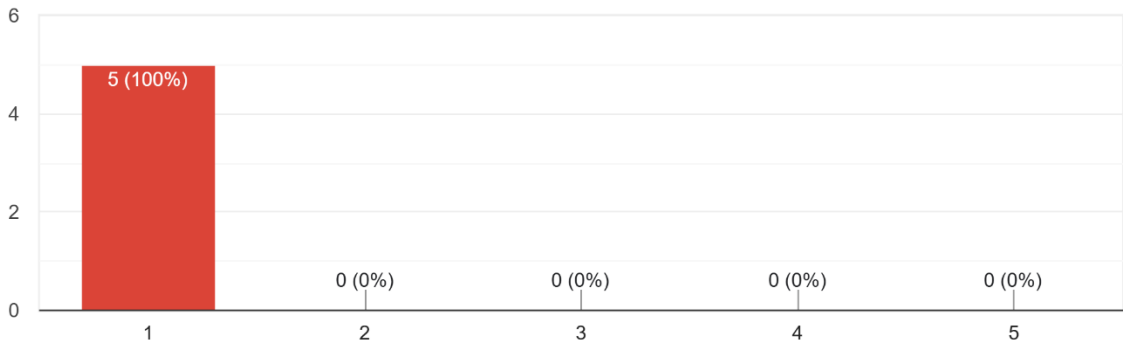
I thought the app was easy to use

5 responses



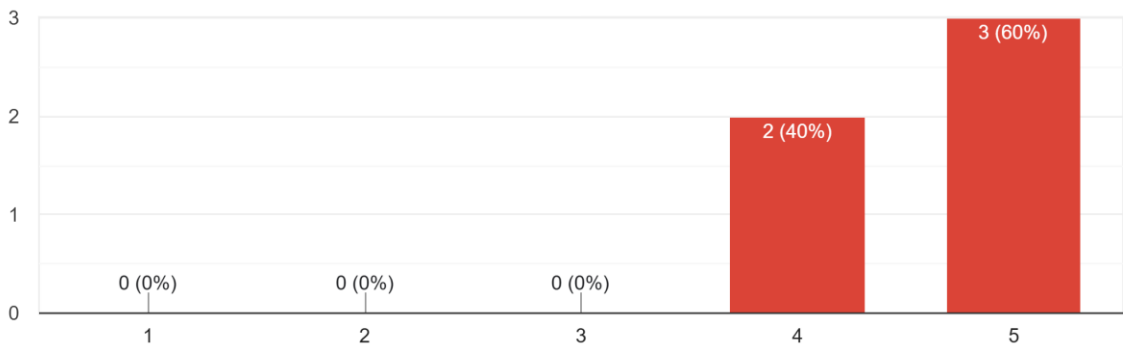
I think that I would need the support of a technical person to use this app

5 responses



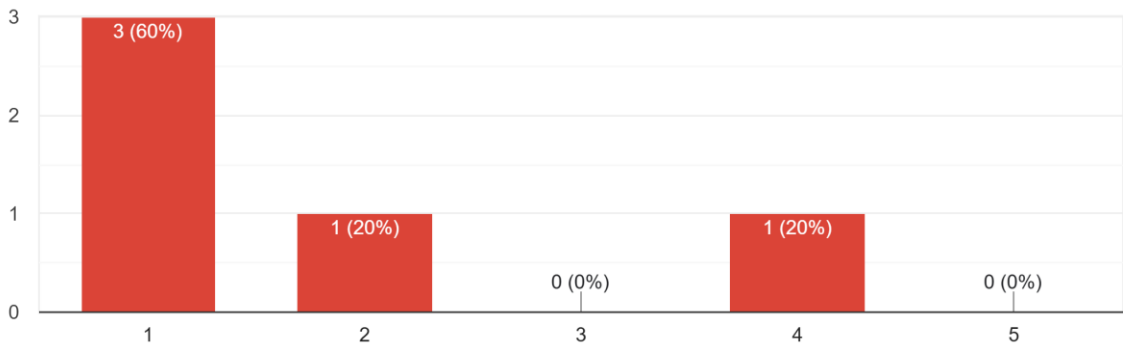
I found the various functions in this app were well integrated

5 responses



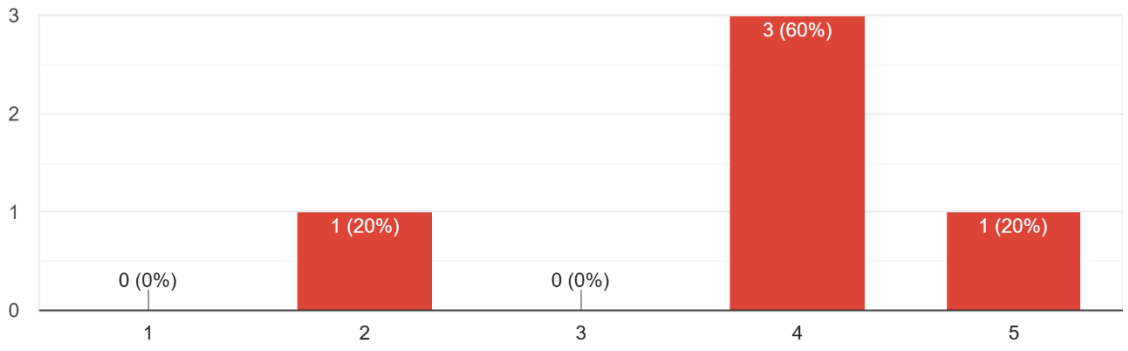
I thought there was too much inconsistency in this app

5 responses



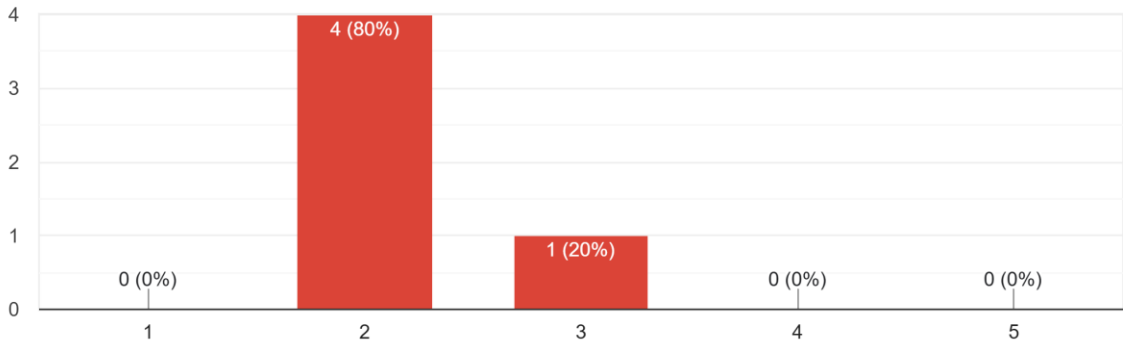
I would imagine that most people would learn to use this app very quickly

5 responses



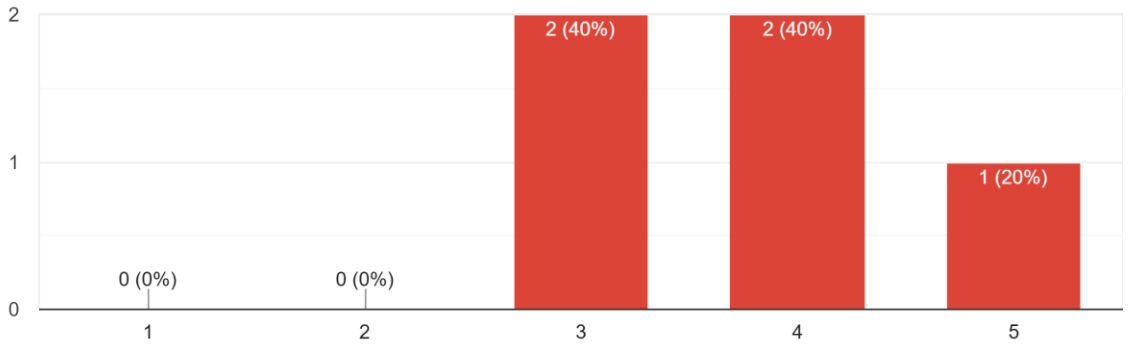
I found the app very cumbersome to us

5 responses



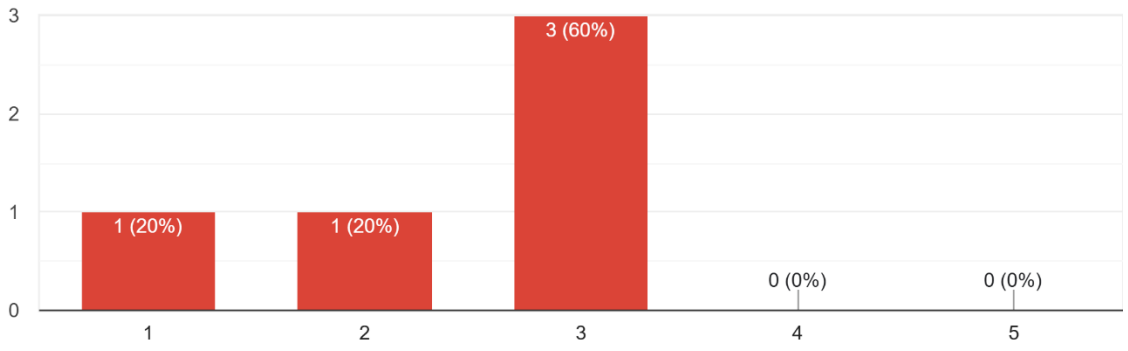
I felt very confident using the app

5 responses



I needed to learn a lot of things before I could get going with this app

5 responses



Section 2: Qualitative Feedback

What was the single most frustrating or confusing moment you experienced with the app, either today or during the week? If none, write "None".

5 responses

- Shuffling through exercises, I feel like it could be presented better than in a huge list
- Navigating the exercise page when I wanted to swap an exercise out.
- Honestly its pretty good I have almost no notes just keep working on it its a really great start
- Setting up the splits was a little annoying mostly because the exercise list is hard to navigate
- Having to manually re-type my weight and reps for every single set. It should remember what I did last time.

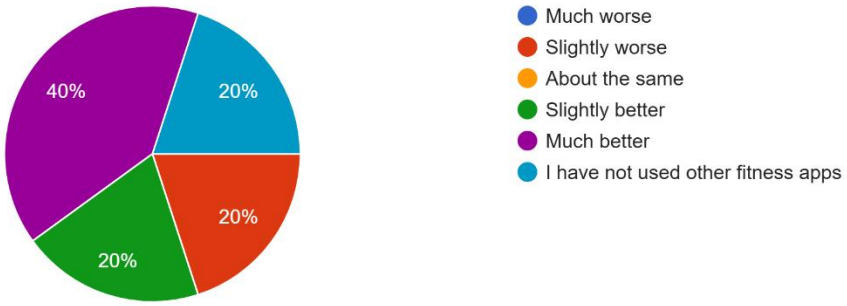
What was the single best or most useful feature of the app?

5 responses

- PR tracking or badges theyre both great I really loveed it
- The badges. Getting the 7-day streak notification made me feel like I'd actually accomplished something.
- The attendance tracking and heatmaps. Seeing the green boxes for my week is very satisfying.
- the logging and attendance mapping
- Freestyle sessions. The flexibility to log a random workout without a pre-set routine is great for busy days.

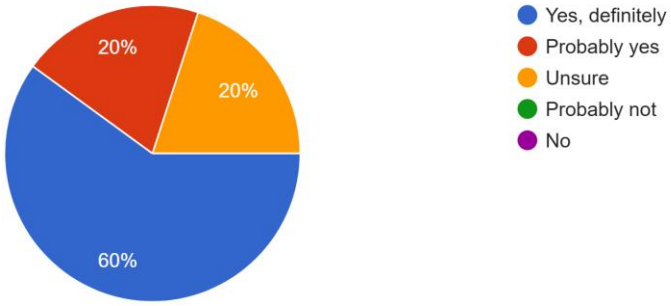
Compared to any fitness apps you have used before, how does OneRep compare?

5 responses



Would you continue using this app after this study?

5 responses



Any other comments or suggestions?

5 responses

More development time could genuienly make this flawless please dont stop and upload to the play store soon

please let me customize the UI. I'd love to change the theme colors to match my gym gear or just to make it "mine."

Maybe some simple tutorials or "first-time" pop-ups to explain the terminology.

great app real easy to use thanks for inviting me to test!

implement auto-fill for the last session's data. If I'm doing a program, I want to see my previous targets immediately.

Appendix G — Raw SUS Scores Table

The full anonymised response data (P1–P5, all 10 SUS items). This supports Table 6.4 in the main report.

Table G.1 — Raw SUS item responses by participant (1 = Strongly Disagree, 5 = Strongly Agree)

Item	Statement	P1	P2	P3	P4	P5
Q1	I think that I would like to use this app frequently	4	3	4	5	5
Q2	I found the app unnecessarily complex	2	2	2	1	1
Q3	I thought the app was easy to use	4	4	5	5	5
Q4	I think that I would need the support of a technical person to use this app	1	1	1	1	1
Q5	I found the various functions in this app were well integrated	5	5	4	4	5
Q6	I thought there was too much inconsistency in this app	1	4	1	2	1
Q7	I would imagine that most people would learn to use this app very quickly	5	4	4	4	2
Q8	I found the app very cumbersome to use	2	2	3	2	2
Q9	I felt very confident using the app	3	3	4	4	5
Q10	I needed to learn a lot of things before I could get going with this app	3	3	2	1	3

Table G.2 — SUS score calculation by participant

	P1	P2	P3	P4	P5
Converted sum (odd items −1, even items 5−score)	32	27	32	35	34
SUS score (sum × 2.5)	80.0	67.5	80.0	87.5	85.0
Grade	B	C	B	B+	B+

Mean SUS score: 80.0

Scoring method: Brooke (1996). Grade boundaries: Bangor, Kortum and Miller (2008).

Appendix H — Think-Aloud Observation Notes

Your observation sheets from all five sessions. These support the quotes and findings in Section 6.4. They do not need to be polished — raw notes are fine and actually more credible.

These notes were recorded by the researcher during in-person think-aloud sessions conducted following the seven-day naturalistic usage period. Notes are reproduced as written at the time of the session. Participants are identified by pseudonym only.

H.1 — P1 *Gym experience: 3–5 years | Prior app: Hevy (briefly) | Current method: Notes app*

Task	Completed	Approx. time	Hesitations / errors	Think-aloud comments
T1 — Create split and routine	Y	1m 05s	None	"Okay, 'New Split'… naming it 'Test Split'. Save. Right, now 'Push Day'… that was pretty instant."
T2 — Add exercise, log set	Y	1m 30s	Scrolling through exercise list to reach B	"Adding Bench… scrolling down to B… there it is. Start. First set, let's say 60 for 10. Done. The UI is clean, not cluttered."
T3 — Start rest timer	Y	10s	None	"Oh! It just started. I didn't even have to look for it. That's actually great because I usually just guess when I'm using my Notes app."
T4 — Progress screen	Y	40s	None	"Progress tab… okay, I see my streak. 4 days. It's simple, I like that I don't have to dig for it."
T5 — Badges screen	Y	25s	None	"Badges… alright, I've got the 'First Workout' one. The others are locked… makes me want to finish my week."

Post-task responses: Q1 (confusion/frustration): "None, really. It felt very natural." Q2 (what worked well): "The rest timer. It stops me from getting distracted on my phone between sets." Q3 (changes/additions): "Keep it as it is. Don't add a bunch of useless features."

Researcher observations: P1 moved with high confidence throughout. No visible learning curve. Treated the app as a direct, improved replacement for their Notes app workflow. Rest timer reaction was spontaneous and unprompted — the strongest single validation of FR-06 in the study.

H.2 — P2 *Gym experience: 1–3 years | Prior app: Strava (regularly) | Current method: Dedicated fitness app*

Task	Completed	Approx. time	Hesitations / errors	Think-aloud comments
T1 — Create split and routine	Y	2m 05s	Brief pause — looked for routine creation on split list screen before tapping into the split	"New split… name it 'Test Split'. Wait, how do I add the routine? Oh, I have to click into the split first. Got it. 'Push Day'."
T2 — Add exercise, log set	Y	2m 50s	Extended scrolling through exercise list, audible sighing observed	"Add exercise. Bench press… B… scrolling… still scrolling… wow, this is a long list. Still scrolling. Okay, found it. Logged. 80kg."
T3 — Start rest timer	Y	15s	None	"Timer is going. Cool."
T4 — Progress screen	Y	55s	Paused on heatmap and asked if it could be shared	"Progress… heatmap looks cool. Can I share this? I'd like to send this to my gym group chat."
T5 — Badges screen	Y	40s	None	"I have the '10 PBs' badge. Nice. I'm competitive, so I like seeing these."

Post-task responses: Q1 (confusion/frustration): "Finding the exercise was annoying. I felt like I was scrolling forever just for a basic bench press." Q2 (what worked well): "The speed of starting a session once it's set up. It's just 'tap, tap, go'." Q3 (changes/additions): "I want a leaderboard. I want to see how my bench ranks against my friends."

Researcher observations: P2 exhibited visible frustration (heavy sighing) during T2 exercise list scrolling. Strong motivation from social comparison and competitive elements — the leaderboard request was the most specific feature request of any participant. Interest in sharing the heatmap suggests social features would be highly valued by this user type.

H.3 — P3 *Gym experience: 5+ years | Prior app: Trainerize (regularly) | Current method: Dedicated fitness app*

Task	Completed	Approx. time	Hesitations / errors	Think-aloud comments
T1 — Create split and routine	Y	1m 40s	None	"Setting up 'Test Split'. Routine is 'Push Day'. The hierarchy makes sense to me."
T2 — Add exercise, log set	Y	3m 05s	Paused after adding exercise, looked for previous session data before logging	"Okay, Bench Press. Added. Now I'm logging… wait, it doesn't show me what I did last time? I have to remember my weight? That's a bit of a let-down."
T3 — Start rest timer	Y	20s	None	"Timer's fine. It's there."
T4 — Progress screen	Y	1m 10s	Examined PR progression chart in detail	"Looking at my charts. PR progression is clear. I like the volume tracking."
T5 — Badges screen	Y	50s	None	"Badges are a nice touch, but I'm more interested in the data."

Post-task responses: Q1 (confusion/frustration): "Why doesn't it auto-fill? If I'm doing 120kg on leg extensions, I don't want to type '1-2-0' every single week." Q2 (what worked well): "Freestyle sessions. Sometimes I just want to hit a quick workout without a plan, and this handles that perfectly." Q3 (changes/additions): "It needs to pull the data from my last session automatically."

Researcher observations: P3 is the most technically critical participant and the only regular Trainerize user in the cohort — the highest credibility reference point for professional-grade app comparison. The auto-fill absence was raised three times across all data collection phases (Day 3 check-in, think-aloud, post-session form), making it the most consistently reported individual gap in the study. Praise of freestyle sessions was the most emphatic positive response of any participant across any feature.

H.4 — P4 *Gym experience: 1–3 years | Prior app: None (paper journal) | Current method: Paper journal*

Task	Completed	Approx. time	Hesitations / errors	Think-aloud comments
T1 — Create split and routine	Y	3m 30s	Hesitated on routine creation — tapped split name before finding the entry point	"New… Split… okay. 'Test Split'. Now a routine? Uhh… oh, tap the name. Right. 'Push Day'. Save."
T2 — Add exercise, log set	Y	4m 00s	Scrolled through exercise list for an extended period	"Searching for Bench Press. It's… down here somewhere? Scroll… scroll… scroll… okay. Adding set. Typing 40. This is actually much faster than writing it in my book."
T3 — Start rest timer	Y	35s	Brief surprise at automatic notification	[Timer buzzes] "Oh! Scared me for a second. That's great though, I usually lose track of time."
T4 — Progress screen	Y	2m 10s	Paused to examine attendance heatmap beyond task requirement	"Progress… oh look at that. The little green boxes for the days I went. I like that a lot."
T5 — Badges screen	Y	1m 20s	None	"I got a gold one! 'First Workout'. That's rewarding."

Post-task responses: Q1 (confusion/frustration): "Just getting used to where things are. The list of exercises is quite long to look through." Q2 (what worked well): "The ease of logging. I thought an app would be harder than paper, but this is actually easier." Q3 (changes/additions): "Sorting the exercises alphabetically would help me find things faster."

Researcher observations: P4 showed the highest delight response of any participant when receiving the rest timer notification and First Workout badge. Transition from paper journal to app engagement within seven days is the strongest accessibility indicator in the study. The specific request for alphabetical sorting is the most actionable design suggestion from any participant and directly addresses H12.

H.5 — P5 *Gym experience: 6–12 months | Prior app: None | Current method: Notes app*

Task	Completed	Approx. time	Hesitations / errors	Think-aloud comments
T1 — Create split and routine	Y	2m 15s	None	"New Split. 'Test Split'. Routine 'Push Day'. Easy."
T2 — Add exercise, log set	Y	2m 40s	Scrolled through exercise list	"Adding Bench Press. I wish this was sorted by body part or something. Scrolling… scrolling… okay, got it. Logged."
T3 — Start rest timer	Y	15s	None	"The timer is nice. It keeps me on track."
T4 — Progress screen	Y	1m 00s	None	"Streak of 7! I've been really consistent this week because of the app."
T5 — Badges screen	Y	35s	None	"I love the badges. I'm trying to get the 30-day one next. It makes the gym feel like a game."

Post-task responses: Q1 (confusion/frustration): "Navigating the exercise page is a bit of a chore." Q2 (what worked well): "The attendance tracking. Seeing my streak grow is the only reason I went to the gym on Tuesday." Q3 (changes/additions): "Let me customize it. I want to change the colours or have a widget on my home screen."

Researcher observations: P5 is the target average gym-goer — lowest experience, no prior app, Notes app background. Highest gamification engagement of any participant. The statement "the only reason I went to the gym on Tuesday" is the single most significant qualitative data point in the study. Completed all tasks with high confidence despite being the least experienced user — strongest indicator of learnability.

Appendix I — Heuristic Evaluation Working Notes

The following notes document the researcher's heuristic evaluation of OneRep conducted prior to participant testing. The evaluation was conducted against Nielsen's (1994) ten usability heuristics. Each screen was inspected systematically. Severity ratings follow Nielsen's (1994) scale: 0 = not a problem, 1 = cosmetic, 2 = minor, 3 = major, 4 = catastrophic.

I.1 — Screen-by-Screen Notes

Splash Screen / AuthScreen H1: No loading indicator present during app initialisation or sign-in. On slower devices the app could appear frozen for 1–2 seconds. Severity 1 — cosmetic, unlikely to cause abandonment. Not resolved — low priority. H9: Sign-in error message reads "Invalid credentials" regardless of whether email or password is incorrect. Does not help user identify which field to correct. Severity 2. Not resolved within project timeline.

HomeScreen (initial state) H7: Empty states on Splits tab display passive text ("No splits yet") without a call to action directing the user toward the FAB. A new user has no prompt telling them what to do first. Severity 2. Not resolved — would require UI iteration. H11: No onboarding flow, tutorial, or first-time guidance. A first-time user arriving at an empty SplitListScreen receives no contextual help. Severity 2. Not resolved — identified as future work.

SplitListScreen H3: Swiping a split card soft-deletes it immediately with no undo snackbar or confirmation dialog. A user who accidentally swipes will not be able to recover their split without re-creating it. Severity 3. Not resolved — undo snackbar would be the appropriate fix.

ActiveSessionScreen H4: Pressing the back button or navigating away from an active session produced no warning. All logged sets would be silently lost. Severity 4 — catastrophic. RESOLVED before testing: a "Discard session?" confirmation dialog was added.

ProfileScreen H5: The bodyweight input field accepted negative values and non-numeric characters without validation. Severity 2. RESOLVED before testing: input validation added. H6: No warning presented when clearing profile data (bodyweight, sex) or when signing out. For a user who has not yet synced, sign-out triggers clearLocalData() which wipes all local records permanently. Severity 4 — catastrophic. RESOLVED before testing: confirmation dialog added to sign-out flow.

RoutineDetailScreenH8: No mechanism to reorder exercises within a routine. Users who want to organise their routine cannot do so within the app. Severity 2. Not resolved

— drag-to-reorder identified as future work. H12: Exercise library is presented as a single unsorted flat list of 40+ items. No alphabetical sorting, no search, no category filter. Finding a specific exercise requires full-list scrolling. Severity 3. Not resolved within project timeline — this was the most significant pre-testing risk.

ExerciseDetailScreen H9: When bodyweight and sex had not been entered in the profile, the strength percentile widget left a visible white gap in the screen layout rather than hiding cleanly. Severity 1 — cosmetic. RESOLVED before testing: widget replaced with a nudge card when profile is incomplete.

ProgressScreen No violations identified beyond those already noted for empty states (H7 above).

BadgesScreen No violations identified. Earned/unearned distinction is visually clear. Badge descriptions are informative.

I.2 — Summary of Pre-Test Resolutions

Before participant testing commenced, the following violations were resolved in a rapid iteration cycle:

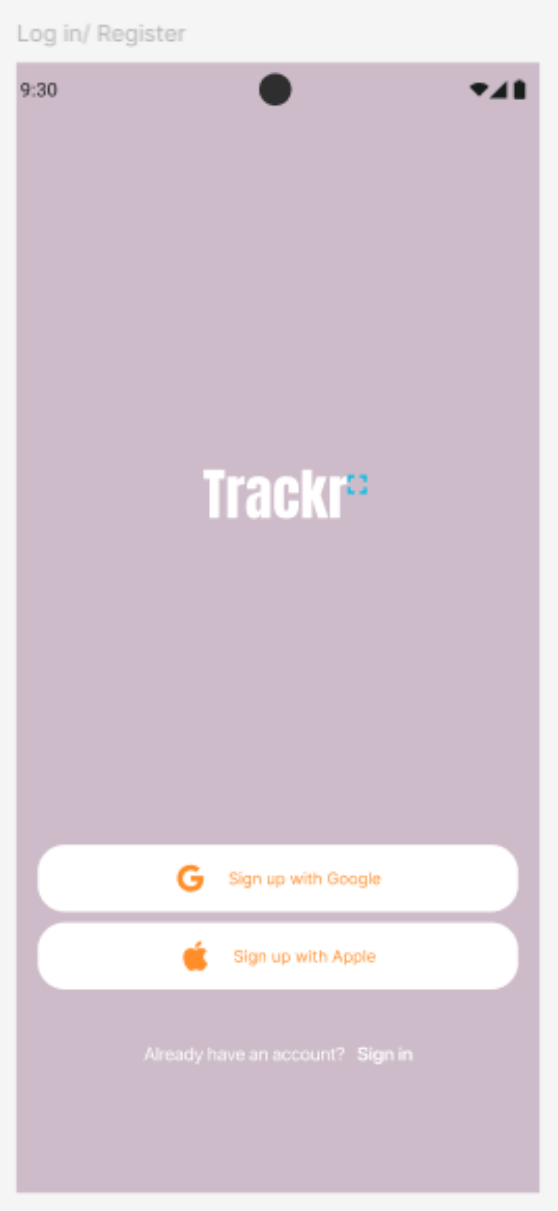
Violation	Fix Applied
H04 — No session discard confirmation	Added "Discard session?" confirmation dialog on back navigation
H05 — Bodyweight field accepts invalid input	Added numeric-only input validation with range check
H06 — No sign-out warning	Added "Sign out and clear local data?" confirmation dialog
H02 — Sync icon provides no feedback	Added CircularProgressIndicator during sync and SnackBar on completion
H09 — Percentile widget white gap	Replaced with nudge card when profile data is absent

I.3 — Violations Carried Into Testing

The following violations were identified but not resolved before participant testing, either due to time constraints or because their severity did not warrant delaying the testing phase:

Violation	Rationale for deferral
H01 — No splash loading indicator	Severity 1 — cosmetic, unlikely to affect study results
H03 — No undo on split deletion	Severity 3 — noted as testing risk; no accidental deletions occurred during sessions
H07 — Passive empty states	Severity 2 — participants were guided by task instructions, minimising impact
H08 — No exercise reordering	Severity 2 — not tested in task scenarios
H10 — Generic error message	Severity 2 — not triggered during task scenarios
H11 — No onboarding guidance	Severity 2 — mitigated by seven-day familiarisation period
H12 — Unsorted exercise list	Severity 3 — highest pre-testing risk; confirmed by four of five participants

Appendix J - Figma wireframes



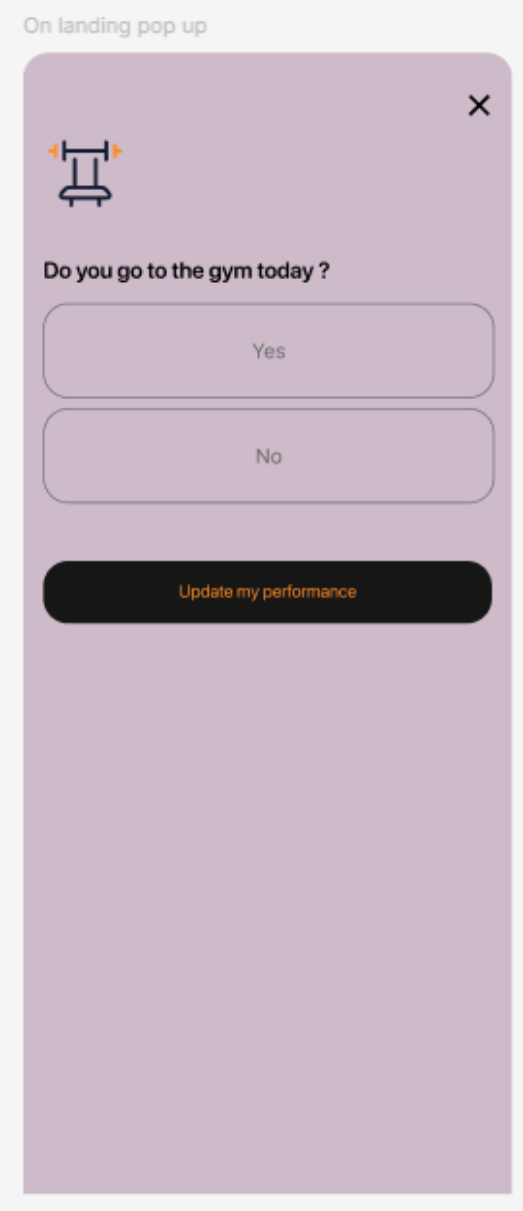
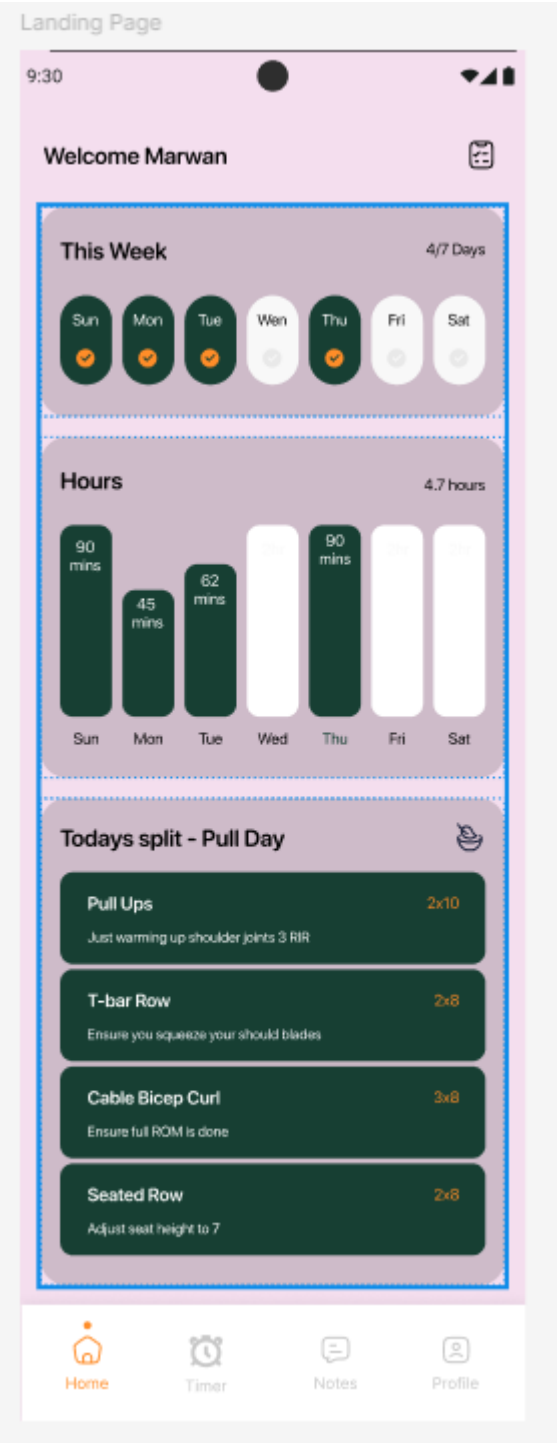


Figure J.1 (Left) - Login/register page.

Figure J.2 (Right) - Gym attendance popup.



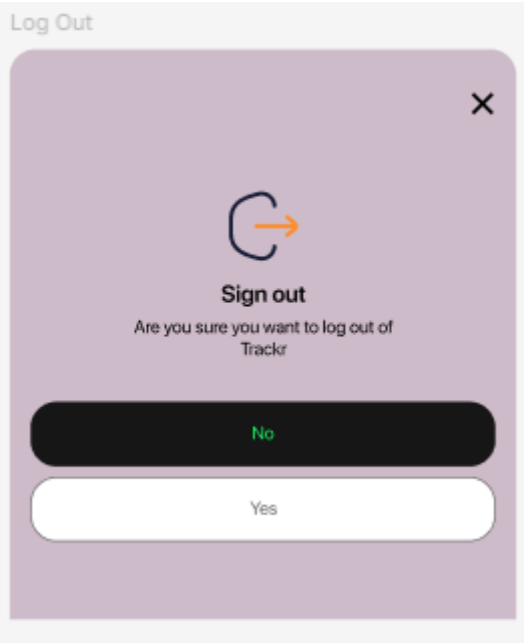


Figure J.3 (Left) - Landing page.

Figure J.4 (Right) - Logout page.

Appendix K — WeChat and Meta Cognitive Walkthrough

K.1 - Meta and the Failure of Integration

Meta's ecosystem presents a contrasting case. Facebook, Instagram, WhatsApp, and Messenger collectively span social networking, communication, content consumption, and commerce — a breadth of functionality comparable to WeChat's. Yet these services remain architecturally and experientially separate. Each application maintains its own navigation logic, feature hierarchy, and interaction patterns, requiring users who wish to accomplish related tasks across Meta's portfolio to switch between applications repeatedly.

This fragmentation imposes precisely the costs identified in Section 2.4. Each application switch forces re-orientation to a new interface, reconstruction of task context, and reliance on recall rather than recognition to locate functionality (Norman, 2013). Meta has pursued partial integration at the technical level — unified messaging infrastructure, cross-posting between platforms — but these efforts operate at the backend rather than the experience level. From the user's perspective, interactions remain divided, and end-to-end service journeys are unsupported (Stickdorn and Schneider, 2012).

The distinction between WeChat and Meta crystallises the argument that motivates this project: aggregation of functionality across multiple applications is not equivalent to integration of functionality within one. Without deliberate system-level coherence and journey-oriented design, a multifunctional ecosystem may exacerbate fragmentation rather than resolve it. This principle applies directly to the fitness domain: a gym-goer using separate applications for workout logging, rest timing, and progress tracking is experiencing Meta's problem at a smaller scale. OneRep is designed to apply WeChat's principle — unified, journey-coherent functionality within a single persistent environment — to that specific use case.

The Fragmented Meta Journey

User Goal: Meet a friend, pay via bank transfer, and post a photo. Tools: WhatsApp (Messaging), Banking App (Payment), Instagram (Sharing).

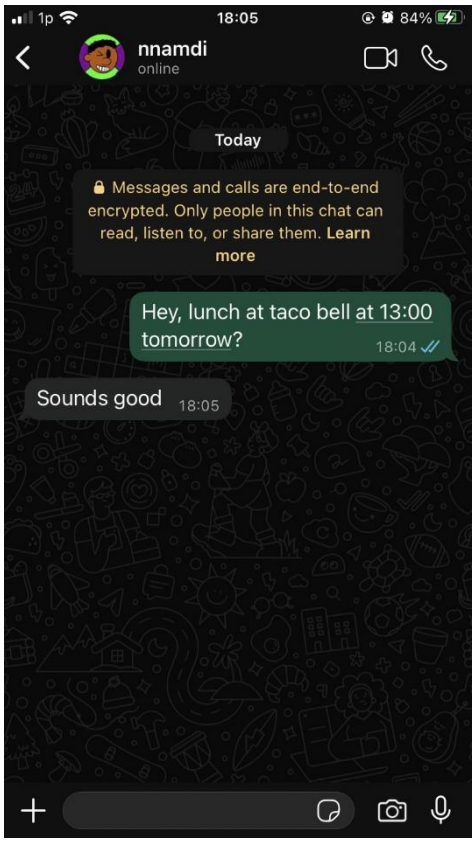


Figure K.1 – Whatsapp messaging a friend.

Step 1: Planning via WhatsApp

- Action: The user sends a message to a friend to coordinate the location.
- Will the user try to achieve the right effect? Yes, the intent is clear (communication).
- Will the user notice the action is available? Yes, the message box is prominent.
- Will the user associate the action with the effect? Yes.
- Success/Progress: The friend replies.
- The "Meta" Friction: To find a restaurant, the user likely has to leave WhatsApp to use Google Maps or Yelp, then copy/paste the address back into the chat. Meta lacks an integrated "Discovery" layer within the chat UI.

Step 2: Paying via Bank Transfer (The Breakpoint)

Figure K.2 – Switching from WhatsApp to a banking app.

- Action: The user minimizes WhatsApp, opens a separate Banking App, authenticates (FaceID/Passcode), enters the friend's details, and sends money.
- Will the user try to achieve the right effect? Yes, but they must now manage external data (the friend's bank info or phone number).
- Will the user notice the action is available? No. Within the Meta ecosystem (WhatsApp), there is no visual affordance or "nudge" to pay unless the user is in a specific region (India) with WhatsApp Pay enabled. From a user experience perspective, region-dependent functionality further undermines predictability and learnability, as users cannot form stable expectations about system capabilities.
- Will the user associate the action with the effect? No. The "Pay" action is mentally associated with the Bank, not Meta.
- Success/Progress: The user sees a confirmation in the Banking app.
- The "Meta" Friction: This is a total departure from the ecosystem. Meta loses all data/context of the transaction. There is no "Transaction Successful" card in the WhatsApp chat to close the loop for both parties.

Step 3: Posting via Instagram

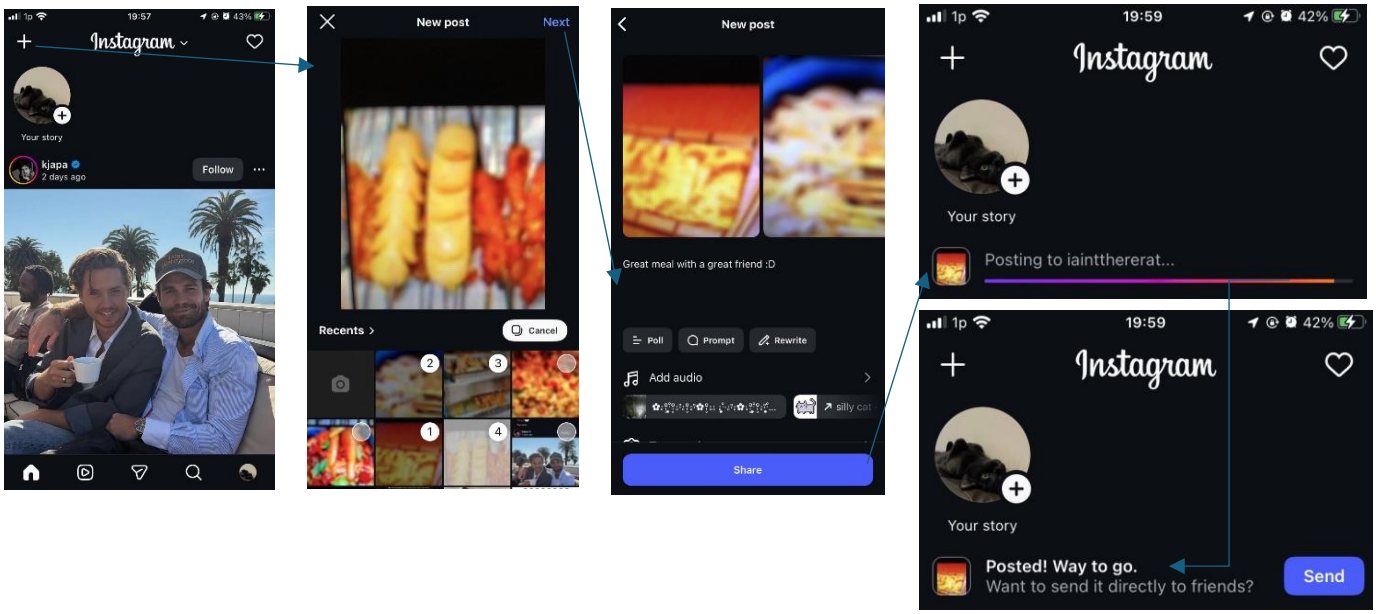


Figure K.3 – Journey of posting an image to Instagram.

- Action: The user closes the banking app (or WhatsApp), opens Instagram, taps the "Plus" icon, selects a photo, applies filters, and posts.
- Will the user try to achieve the right effect? Yes, for social validation/memory.
- Will the user notice the action is available? Yes, the "Post" button is a primary UI element.
- Will the user associate the action with the effect? Yes.
- Success/Progress: The photo appears on the feed.
- The "Meta" Friction: The photo is "trapped" in Instagram. If the user wants to ensure the friend they just ate with sees it, they often have to cross-post or send the Instagram link back to the WhatsApp thread.

Failure Type	**Analysis of Meta's Fragmented Journey**
Gulf of Execution	High. The user has to manually move data (Account numbers, URLs, Photos) between three distinct app environments.
Gulf of Evaluation	High. There is no single "State of Truth." To see if the plan is complete, the user must check three different notification centres.

Failure Type	Analysis of Meta's Fragmented Journey
Context Switching	Severe. Each app switch requires a new "load time" in the user's brain, increasing the chance of distraction (e.g., getting sidetracked by an IG Reel while trying to post).

K.2 - WeChat as a Unified Experience

WeChat is the most frequently cited example of successful superapp design. Originating as a messaging application, it has expanded into an integrated ecosystem supporting communication, payments, social networking, content consumption, booking services, and third-party applications via embedded “mini-programs” — all within a single interface.

From a user journey perspective, WeChat’s principal achievement is enabling multi-step tasks — scheduling events, transferring money, accessing services — without requiring the user to leave the platform. This consolidation directly reduces the context-switching costs described by González and Mark (2004). Critically, WeChat’s integration extends beyond feature aggregation: the platform maintains consistent interaction patterns, navigation structures, and visual language across its services, enabling users to transfer knowledge from one function to another. This favours recognition over recall (Norman, 2013) and preserves the sense of continuity that fragmented ecosystems destroy.

WeChat also demonstrates that multifunctional design need not overwhelm users. Services are surfaced contextually in response to user intent rather than presented simultaneously, a pattern consistent with the brief, goal-oriented interaction bursts that characterise effective mobile engagement (Oulasvirta et al., 2005). It must be acknowledged, however, that WeChat’s dominance is partly attributable to its cultural and regulatory context — a market in which centralised digital services are expected and alternatives are limited. The design principles it demonstrates are transferable; the scale of its adoption is not.

The Integrated WeChat Journey

1. Step: Messaging (The Coordination Phase)

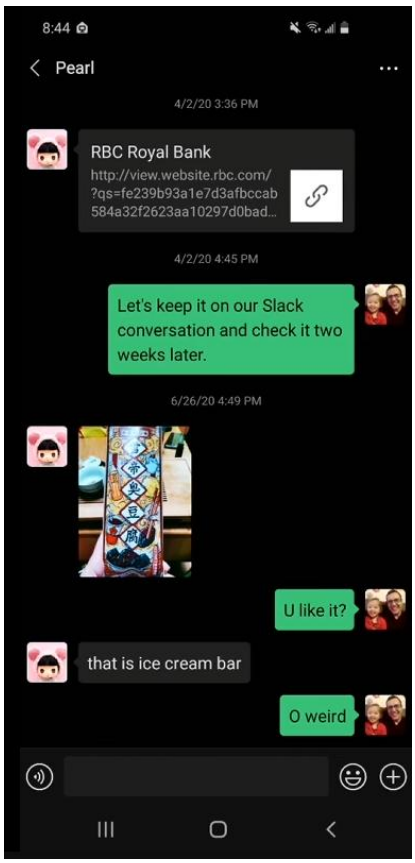


Figure K.4 - Messaging on WeChat.

- **Action:** User opens a chat with a friend and sends a message asking them to lunch.
- **Success Criterion 1 (Correct Effect):** Yes, the goal is to communicate.
- **Success Criterion 2 (Action Visibility):** Yes, the text input and photo-sharing icons are permanent fixtures.
- **Success Criterion 3 (Association):** Yes, standard messaging mental model.
- **Success Criterion 4 (Feedback):** Outgoing message bubbles and "read" indicators (if enabled).

2. Step: Red Packet (The Integrated Payment)

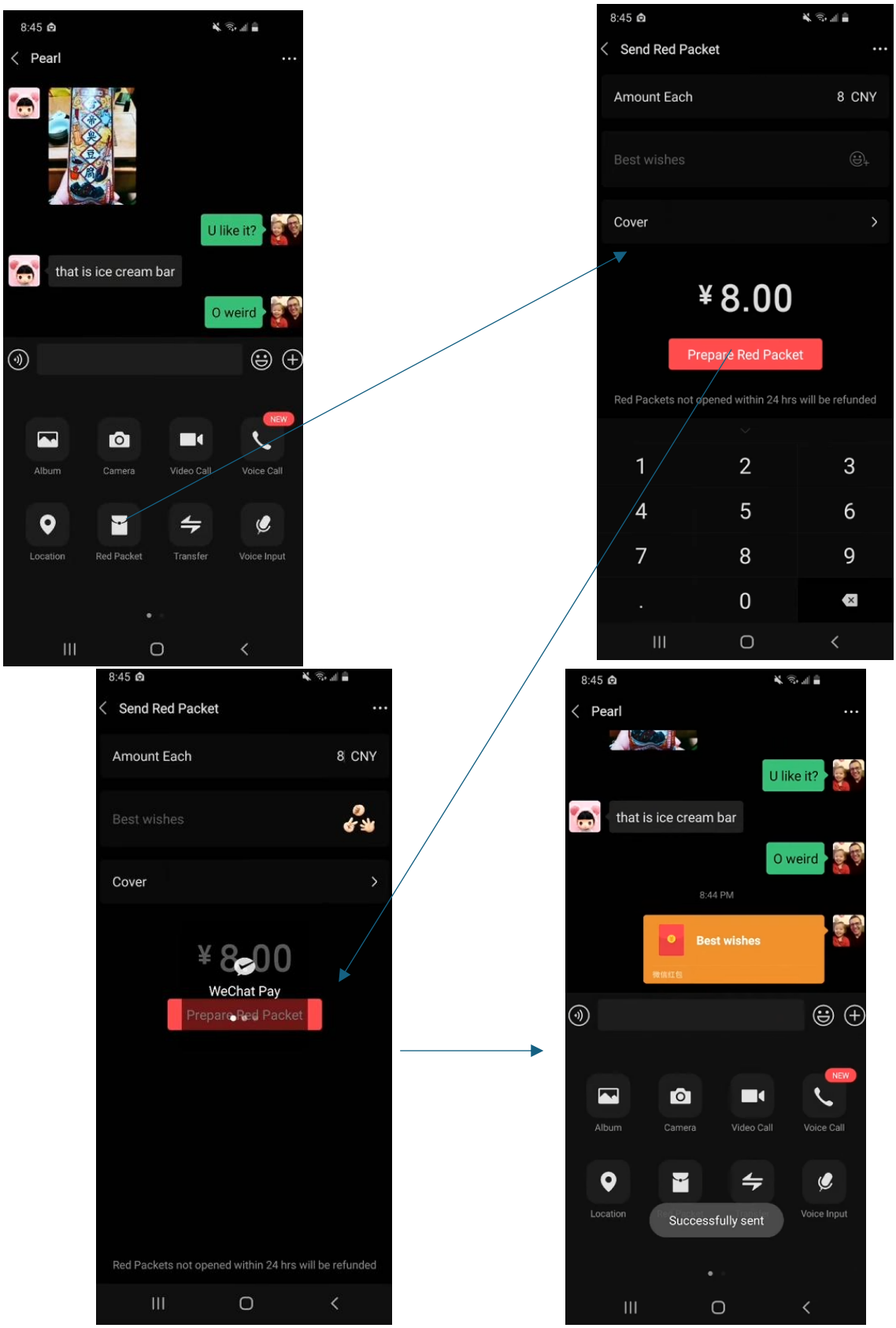


Figure K.5 – Paying a friend on WeChat.

- **Action:** User taps the **"+"** menu in the chat, selects **"Red Packet"**, enters the amount, and sends it directly into the chat thread.
- **Success Criterion 1 (Correct Effect):** Yes. Instead of a "Bank Transfer," the user views this as a "Gift" or "Settling up" within the social context.
- **Success Criterion 2 (Action Visibility):** High. It is a top-level icon in the chat's expansion menu.
- **Success Criterion 3 (Association):** Very High. In WeChat's ecosystem, "paying a friend" is synonymous with "Red Packet."
- **Success Criterion 4 (Feedback): Instant and Dual.** A bright orange envelope appears in the chat. When the friend "opens" it, the user receives a notification that the money has been collected.
- **Cognitive Insight:** Unlike the Meta journey, where the user must leave to a banking app, the Red Packet keeps the "debt" and "repayment" visible to both parties in the conversation history.

3. Step: Uploading to Moments (The Narrative Phase)

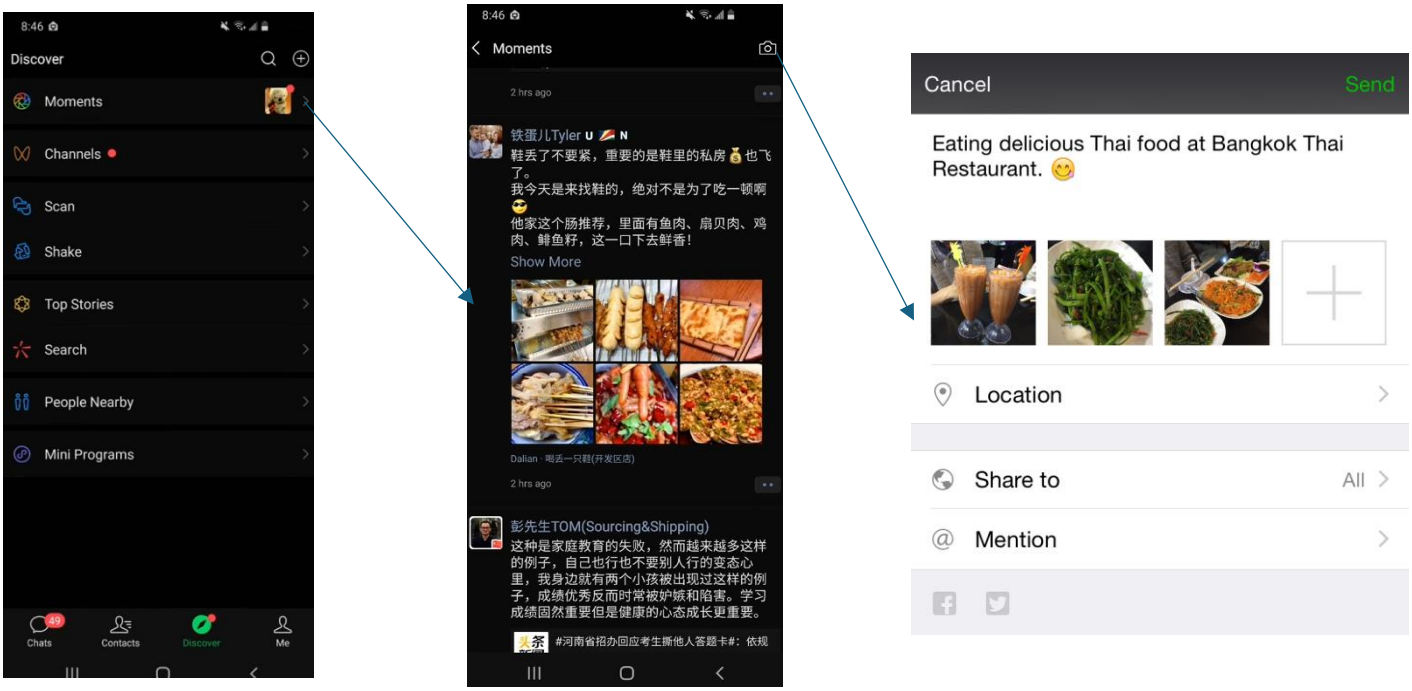


Figure K.6 – Uploading a post on WeChat.

- **Action:** User navigates to **Discover > Moments**, taps the camera icon, selects the meal photos, and tags the friend.
- **Success Criterion 1 (Correct Effect):** Yes, to document the day.

- **Success Criterion 2 (Action Visibility):** Yes, the Moments tab is a primary navigation point.
- **Success Criterion 3 (Association):** Yes, the camera icon is the universal symbol for "Post."
- **Success Criterion 4 (Feedback):** The post appears at the top of the timeline; likes/comments from the friend appear in the "Discover" tab notification.
- **Cognitive Insight:** Because the friend is already the person the user just messaged and paid, the "Social Graph" is identical. There is zero friction in finding the right person to tag or share with.

Executive Summary: The Architecture of User Flow

The fundamental difference between WeChat and Meta is not the availability of features, but the **continuity of the state**. In WeChat, the user is a single entity moving through different modes of an ecosystem. In Meta, the user is an "integrator" forced to manually bridge the gaps between distinct business units. It should be noted that WeChat's effectiveness is partly enabled by cultural norms that favour platform centralisation; however, the interaction patterns themselves remain transferable as design principles.

The "Highs" of WeChat: The Encapsulated Super-App

WeChat succeeds by reducing the **Interaction Cost** to its theoretical minimum.

- **Unified Identity & Social Graph:** Because the messaging identity is the same as the payment and social media identity, there is zero friction in "finding" the context or person for the next step.
- **The "In-Situ" Loop:** By integrating payments (Red Packets) and services (Mini Programs) directly into the chat interface, WeChat eliminates the **Gulf of Evaluation**. Feedback is immediate and shared between both parties in the conversation.
- **Zero-Exit Journey:** The "Mini Program" architecture allows the user to perform complex tasks (booking, ordering, paying) without ever seeing the phone's home screen. This prevents **Inattentional Blindness** and distraction.

The "Lows" of Meta: The Fragmented Federation

Meta's failure to create a unified journey is a result of **Architectural Redundancy** and lack of shared context.

- **Context-Switching Penalty:** Every time a user exits WhatsApp to pay via a bank or post on Instagram, they incur a "mental reload" cost. Meta's ecosystem relies on the user's short-term memory to carry data (like account numbers or restaurant names) from one app to the next.
- **Broken Feedback Loops:** Meta apps do not "talk" to each other in real-time. A success in one app (e.g., a payment confirmation) is invisible to the other (e.g., the WhatsApp chat), forcing the user to manually verify and report progress.
- **Feature Redundancy:** By "bolting on" similar features (like Stories and Messaging) to every app in the fleet, Meta violates the **Consistency and Standards** heuristic. Users are often confused about where a specific interaction "belongs."

Final Synthesis: Lessons for the Fitness App Prototype

Based on this walkthrough, your proposed fitness application should prioritize the following design principles to avoid the "Meta Trap":

Principle	Strategic Application
Data Ubiquity	Ensure workout stats are automatically available for social sharing and "earning" rewards without manual exports.
Contextual Payment	If the app involves personal training or challenges, payment should happen *inside* the workout/social thread (the WeChat Red Packet model).
Single Source of Truth	Use one unified notification center so the user never has to "check" multiple tabs to see if their journey (e.g., Workout -> Share -> Reward) is complete.

Appendix L - Privacy Policy for OneRep

Privacy Policy for OneRep

Last updated: April 15, 2026

OneRep is a strength tracking application developed as an independent project. This policy explains what data is collected, how it is used, and your rights as a user.

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1. Data We Collect

When you create an account and use OneRep, the following data is collected:

Data	Purpose
Email address	Account creation and authentication
Password (hashed, never stored in plain text)	Account authentication via Supabase Auth
Workout data (sessions, sets, reps, weights, timestamps)	Core app functionality and cross-device sync
Personal bests (exercise, weight, reps, date achieved)	Tracking your performance records
Workout routines and splits	Saving and restoring your training programmes
Badges earned (achievement keys and dates)	Tracking your in-app achievements
Custom exercises (name, body part, equipment type)	User-created exercise library entries

We do not collect your name, phone number, location, payment information, photos, contacts, or any device identifiers.

2. How Your Data Is Used

Your data is used exclusively to provide the functionality of the app:

- Authenticating your account
- Syncing your workout data across devices
- Restoring your data when you sign in on a new device
- Displaying your personal bests and progress history

Your data is **never used for advertising, profiling, analytics, or sold to third parties.**

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3. Data Storage and Third Parties

OneRep uses **Supabase** (supabase.com) as its backend infrastructure for authentication and data storage. Supabase processes your data on our behalf under their data processing agreements.

- Supabase Privacy Policy: https://supabase.com/privacy
- Your workout data is stored in Supabase's cloud infrastructure and is associated with your user account.
- No other third-party services receive your data.

4. Data Retention and Deletion

- Your data is retained for as long as your account exists.
- When you sign out of the app, all locally stored data is deleted from your device.
- To request permanent deletion of your account and all associated data, contact us at the email address below. We will process deletion requests within 30 days.

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5. Your Rights (UK GDPR)

As a user based in the United Kingdom or European Economic Area, you have the right to:

- **Access** the personal data we hold about you
- **Rectify** inaccurate data
- **Erase** your data ("right to be forgotten")
- **Restrict** processing of your data
- **Data portability** — receive your data in a structured, machine-readable format
- **Object** to processing

To exercise any of these rights, contact us at the email address in Section 7.

6. Security

We take reasonable technical measures to protect your data:

- Passwords are managed by Supabase Auth and are never stored in plain text
- Data in transit is encrypted via HTTPS/TLS
- Access to your data in Supabase is restricted by Row-Level Security policies tied to your authenticated user ID

No system is perfectly secure. If you believe your account has been compromised, sign out immediately and contact us.

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7. Contact

If you have questions about this policy or wish to exercise your data rights, contact:

Email: DiggleShyStudios@protonmail.com

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8. Changes to This Policy

We may update this policy as the app evolves. If material changes are made, we will update the "Last updated" date at the top of this document. Continued use of the app after changes constitutes acceptance of the updated policy.

Appendix M – Google Play Console status

The app is in the final stages of being uploaded to the Play Store. It's currently in closed testing with 7 out of the required 12 tested opted in.

App	Installed audience	App status	Update status	Last updated	
OneRep com.marwan.fitness_app	0	Closed testing		Apr 15, 2026	View app →

Show rows: 10 1 - 1 of 1

Dashboard

Get ready to publish your app. Show more

OneRep

com.marwan.fitness_app

Production

Inactive

Show test tracks

Production

Apply for access to production

Production is where you make your app available to billions of users on Google Play. Before you can apply for production access, you need to run a closed test which meets our criteria. When you apply, you'll also need to answer some questions about your closed test. Preview questions

- ✓ ~~Publish a closed testing release~~
- Have at least 12 testers opted-in to your closed test
  7 tester currently opted-in
- Run your closed test with at least 12 testers, for at least 14 days

Apply for production Learn more

Appendix N - Test descriptions

PersonalBestRepository tests (personal_best_repository_test.dart) verify the core PR detection algorithm across all four metric types and boundary conditions: the first set logged for any repetition count is always recorded as a PR; PRs at different repetition counts are tracked independently; progressive overload across multiple sessions is correctly tracked; and beating a five-repetition PR does not affect the ten-repetition PR record for the same exercise.

BadgeService tests (badge_service_test.dart) verify all eight badge triggers and their associated edge cases: all badges are seeded as unearned after setUp; re-seeding does not overwrite badges already earned; each badge fires at the correct threshold; badge award is idempotent — awarding the same badge twice does not alter earnedAt; and the first_custom_exercise badge correctly distinguishes custom exercises from seeded default exercises.

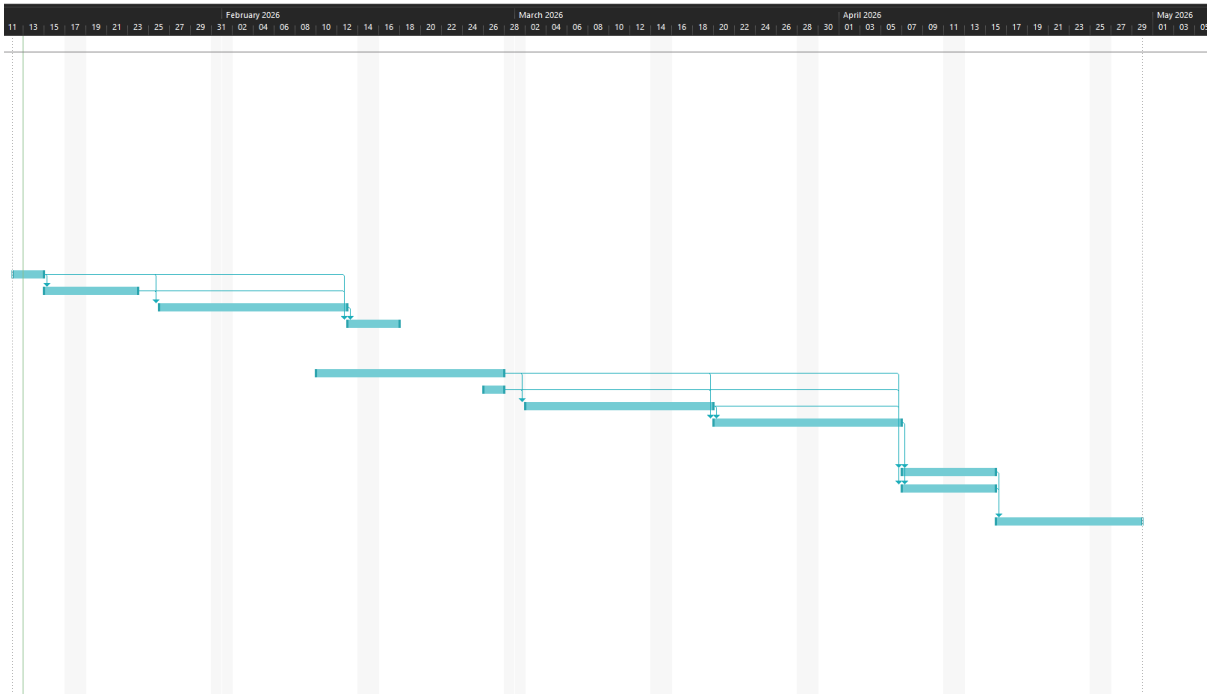
Strength Standards tests (strength_standards_test.dart) verify the calculatePercentile() function across the full input space: hasStrengthStandards returns the correct boolean for supported and unsupported exercises; the function returns null for unknown exercises, zero lifts, and negative lifts; male and female band boundaries at all five threshold levels are correct; bodyweight clamping at the minimum and maximum bracket bounds behaves correctly; the same lift produces different percentile results for male and female users; linear interpolation at the midpoint between two bands produces the arithmetically correct result; and the Overhead Press alias correctly resolves to the same standards table as Shoulder Press.

ExerciseRepository tests (exercise_repository_test.dart) verify basic CRUD operations: insert and retrieve a single exercise; delete an exercise; insert multiple exercises and retrieve all.

SplitRepository tests (split_repository_test.dart) verify the split hierarchy operations: create and retrieve a split; delete a split; add a routine to a split; add an exercise to a routine; deleting a split cascades correctly to its child routines.

SyncService tests (sync_service_test.dart) verify the dirty flag mechanism and sync readiness logic: new inserts have null syncedAt; records with syncedAt set are not dirty; updating a synced record clears syncedAt; newly inserted sessions are dirty; dirty records remain unsynced when userId is null; a routine without a synced parent split is skipped; a routine becomes eligible once its parent split has a remoteId; a set without a synced parent session is skipped; a soft-deleted split has deletedAt set and syncedAt cleared; and soft-deleted records appear in the dirty queue.

Appendix P – Gannt chart and table



Task Name	Duration	Start	Finish	Predecessors
~~Phase 1: Project Definition & Planning~~				
~~Phase 2: Literature Review & Theoretical Foundation~~				
~~Phase 3: Applied Analysis & Case Studies~~				
Phase 4: Requirements Definition & Design Exploration				
Identify core functional requirements				
Map user journeys for fitness use case				
Produce low-fidelity wireframes (Figma)				
Phase 5: Prototype Development				
Select development framework	3 days	Mon 12/01/26	Wed 14/01/26	
Create application skeleton	7 days	Thu 15/01/26	Fri 23/01/26	15
Implement core features (tracking, timers, notes)	14 days	Mon 26/01/26	Thu 12/02/26	15,16
Ensure navigation supports unified journeys	3 days	Fri 13/02/26	Tue 17/02/26	15,16,17
Phase 6: Usability Evaluation				
Design usability testing protocol	14 days	Tue 10/02/26	Fri 27/02/26	
Recruit participants	2 days	Thu 26/02/26	Fri 27/02/26	
Conduct usability testing sessions	14 days	Mon 02/03/26	Thu 19/03/26	21,22
Collect quantitative and qualitative data	12 days	Fri 20/03/26	Mon 06/04/26	21,22,23
Phase 7: Analysis, Refinement & Reporting				
Analyse usability test results	7 days	Tue 07/04/26	Wed 15/04/26	21,22,23,24
Evaluate findings against journey theory	7 days	Tue 07/04/26	Wed 15/04/26	21,22,23,24
Refine prototype (if time permits)				
Finalise project report	10 days	Thu 16/04/26	Wed 29/04/26	27,28